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Case Study: CMS Texas Hospital Quality

Scenario

You are a junior data analyst working for a business intelligence consultant. You have been at your job for six months, and your boss feels you are ready for more responsibility. They have asked you to lead a project for a brand new client. This will involve everything from defining the business task all the way through presenting your data-driven recommendations. You will choose the topic, ask the right questions, identify a fresh dataset and ensure its integrity, conduct analysis, create compelling data visualizations, and prepare a presentation.

Ask

Five questions will guide your case study:

1. What type of company does your client represent, and what are they asking you to accomplish?

-The client is a fictional healthcare quality collaborative that supports hospitals throughout Texas. The collaborative wants to use publicly reported CMS data to understand how Texas hospitals perform in three connected areas: overall hospital quality, patient experience, unplanned hospital readmissions or return visits. Since the client has limited resources and cannot support every hospital or improvement initiative equally, leadership needs an objective way to identify the hospitals and performance areas that appear to require the most attention.

2. What are the key factors involved in the business task you are investigating?

-I am investigating overall hospital quality, patient experience, and unplanned hospital readmissions or return visits. This analysis will focus on hospital level performance rather than individual patients.

3. What type of data will be appropriate for your analysis?

-I will get hospital name, location, type, ownership, emergency services, overall star rating, patients' rating of their care, willingness to recommend the hospital, communication scores, readmission and return-to-hospital measures, and other experience measures.

4. Where will you obtain that data?

-I will combine three official Centers for Medicare & Medicaid services (CMS) datasets:

Hospital General Information, Patient Survey (HCAHPS), and Unplanned Hospital Visits.

5. Who is your audience, and what materials will help you present to them effectively?

-The client is the Texas Hospital Quality Collaborative. They are a (fictional) nonprofit healthcare quality improvement organization that provides benchmarking, reporting, and operational support to hospitals across Texas. I clean, validate, and analyze the data in Excel.

Case Study Roadmap - Ask

Guiding questions

● What topic are you exploring?

-Hospital quality performance, patient experience, and unplanned readmissions among Texas hospitals.

● What is the problem you are trying to solve?

-The Texas Hospital Quality Collaborative lacks a consolidated analysis showing how Texas hospitals perform across overall CMS ratings, patient experience, and unplanned readmissions. Without this analysis, leadership cannot consistently identify the facilities and performance areas that should receive priority quality improvement support.

● What metrics will you use to measure your data to achieve your objective?

-Some planned metrics I will use to examine how Texas hospitals perform are: Total Texas hospitals, average or median overall hospital rating, percentage rated 1-2 stars, percentage rated 4-5 stars, Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) recommendation rate, hospital wide readmission result, percentage worse than national, and priority flag count.

● Who are the stakeholders?

-They are the executive director of the collaborative, chief medical officer, director of quality improvement, patient experience director, data and reporting team.

● Who is your audience?

-The primary audience is the executive leadership and quality improvement committee of the Texas Hospital Quality Collaborative.

● How can your insights help your client make decisions?

-The analysis can help the client direct quality improvement support toward facilities with the greatest apparent needs. It can do this by identifying hospitals with several overlapping performance concerns, identify measures that show statewide weakness rather than insolated hospital weakness, determine whether additional investigation is needed before committing resources, etc.

Key tasks

1. Identify the business task

2. Determine key stakeholders

3. Choose a dataset

4. Establish metrics

Deliverable

- A clear statement of the business task you have selected to investigate

- Which Texas acute care hospitals show the greatest opportunities for improvement based on overall hospital ratings, patient experience results, and unplanned readmission performance?

Prepare

Once you have decided on a dataset, gather and store the data appropriately, clean the data, and make sure it is reliable and error-free. Document your process, as clients often ask to see both raw and cleaned data. Now, prepare your data for analysis using the following Case Study Roadmap as a guide:

Case Study Roadmap - Prepare

Guiding questions

- Where is your data located?

- My data was downloaded from [Hospital General Information | Provider Data Catalog](#), [Patient survey \(HCAHPS\) - Hospital | Provider Data Catalog](#), and [Unplanned Hospital Visits - Hospital | Provider Data Catalog](#) to my local computer.

- How is the data organized?

- The Hospital General Information table is one row per table. The HCAHPS Hospital table is one row per facility and HCAHPS measure. The Unplanned Hospital Visits table is one row per facility and outcome measure. The HCAHPS and unplanned visits files are in long format, so repeated Facility ID's are expected and don't automatically indicate duplicates. The primary key across these three tables is Facility ID. CMS explains that overall rating summarizes measures from mortality, safety, readmission, patient experience, and timely and effective care.

- Are there issues with bias or credibility in this data? Does your data ROCCC?

- There doesn't seem to be bias or credibility issues with this data. The .csv files schema downloaded in tact, and the file names match those in the data dictionary.

Reliable: Strong. CMS publishes standardized hospital quality, patient experience, and readmission data. These files are accompanied by official definitions, reporting cycle information, and footnotes.

Original: Strong. This data comes directly from CMS rather than from a republished Kaggle copy. CMS is the first-party publisher for the Provider Data Catalog datasets used in this project.

Comprehensive: Moderately strong. The files cover than man 5,400 Medicare registered hospitals nationally, however not every hospital has every score, and hospital types may not be directly comparable. The dataset is comprehensive for public CMS reporting, but it is not a complete representation of every aspect of hospital quality.

Current: Strong. The current files were released on 05/13/26 and last modified on 04/13/26. However, measure collection periods vary, and individual scores may represent 1-X numbers of years depending on the measure.

Cited: Strong. Every source has an official dataset page, identifier, data dictionary, release information, and source agency.

● How are you addressing licensing, privacy, security, and accessibility?

-CMS warns that opening CSV files directly in Excel can remove leading zeroes from fields such as provider identifiers, so I will improve them through Power Query during the Process phase. Regarding licensing, works of the US government are generally in the public domain, but [CMS requests attribution as the source and warns that reuse must not imply government endorsement](#). As far as privacy, the files contain public hospital level aggregate information, not identifiable individual patient records. Security-wise, no restricted data is required. For accessibility, the raw CSVs can be opened by common spreadsheet and text tools.

● How did you verify the data's integrity?

-Full validation will happen after importing the data. I have checked that the three files successfully downloaded, retaining their original filenames, schemas, and row count.

● How does it help you answer your question?

-The overall rating is not independent of patient experience and readmission performance. Therefore, I can use all three as separate screening signals, but I cannot describe a total flag count as a validated composite quality score. It will only be a prioritization aid. Together, the sources allow me to analyze which hospitals have low overall ratings, weak patient experience results, and/or unfavorable readmission performance; whether performance differs by ownership or type, patient experience and readmission outcomes appear related; which hospitals display more than one improvement signal, etc.

● Are there any problems with the data?

-Yes. There are lead zero risks, missing and suppressed values, numeric fields stored as text, different table grains, different reporting periods, hospital type comparability, HCAHPS response variation, readmission results may be unavailable, and overall ratings overlap with other metrics.

Key tasks

● Download data and store it appropriately.

● Identify how it's organized.

● Sort and filter the data.

- Determine the credibility of the data.

Deliverable

- A description of all data sources used

-This analysis uses three hospital level datasets from the CMS Provider Data Catalog. Hospital General Information provides facility identifiers, geographic information, hospital type, ownership, emergency service status, and the CMS overall hospital rating. The Patient Survey HCAHPS Hospital table contains standardized patient experience measures, including hospital recommendation, overall patient rating, communication, discharge information, survey volume, and response rate. The Unplanned Hospital Visits Hospital table provides hospital level readmission and returns visit measures, including scores, national performance comparisons, denominators, confidence estimates, reporting dates, and footnotes. The CMS Hospital Data Dictionary was used to interpret field definitions, measure IDs, reporting periods, and unavailable or suppressed results. All sources are publicly available hospital level data and do not contain identifiable patient records.

Process

Case Study Roadmap - Process

Guiding questions

- What tools are you choosing and why?

-I will use Excel Power Query for importing, cleaning, and merging to create the final analytical table. Power Query M code will be used for reproducible code. I will then use Excel worksheets for analysis. I will not use VLOOKUP to combine the three primary datasets. Power Query merges are more appropriate for this project because they create a documented join between related tables based on Facility ID.

- Have you ensured your data's integrity?

-Yes. I preserved the original CMS files by filtering the data through Power Query rather than deleting rows manually. I also separated unavailable values from numeric zeroes, checked candidate keys for duplicates, verified that the merged table remains at the expected grain, checked unmatched records for coverage, and confirmed that the final analysis table has the same row count as the hospital master table.

- What steps have you taken to ensure that your data is clean?

-I selected relevant columns, trimmed key text fields, filtered to Texas, restricted the main population to acute care hospitals, converted valid numerical values while retaining unavailable results as null, pivoted HCAHPS from long to wide format, merged the three cleaned tables by Facility ID, and ran duplicate, row count, join, missing value, and error checks.

● How can you verify that your data is clean and ready to analyze?

-Data integrity was verified through duplicate key checks, identifier checks, numeric range checks, unmatched record reviews, row count reconciliation, data profiling, manual comparisons with original CMS records, and confirming measures aligned with business questions.

● Have you documented your cleaning process so you can review and share those results?

-Yes, in the changelog named cms-tx-hospital-quality-data-cleaning-changelog.pdf.

Key tasks

● Check the data for errors.

● Choose your tools.

● Transform the data so you can work with it effectively.

● Document the cleaning process.

Deliverable

● Documentation of any cleaning or manipulation of data

-All data cleaning and validation has been documented in the file called cms-tx-hospital-quality-data-cleaning-changelog.pdf.

Analyze

Case Study Roadmap - Analyze

Guiding questions

● How should you organize your data to perform analysis on it?

-The merged table uses one row per Texas acute care hospital in tblAnalysisModel. Hospital characteristics, overall rating, selected HCAHPS measures, readmission measures, reporting availability, benchmarks, quartiles, and screening flags appear in the same table. This structure supports formulas, PivotTables, segmentation, and relationship analysis.

● Has your data been properly formatted?

-Yes. One row represents one hospital. Each column is assigned the correct data type. Unavailable results are blank rather than zero. CMS percentage points use 0.0\%. Calculated ratios use normal percentage formatting. Categorical fields have consistent labels.

● What surprises did you discover in the data?

-One of the most notable findings was the difference in data coverage: only 70.2% of hospitals had a reportable overall CMS rating, compared with approximately 90% coverage for HCAHPS and readmission measures. The analysis also showed that hospital quality is multidimensional; several hospitals with four- or five-star overall ratings still had below-median patient recommendation scores, and several high rated hospitals were classified

as worse than national on readmission. Patient recommendation and readmission were inversely associated, but the relationship was moderate rather than strong ($r = -0.301$).

● What trends or relationships did you find in the data?

-The analysis found that Texas acute care hospitals were concentrated primarily in the three- and four-star categories, with 47.0% of rated hospitals receiving four or five stars. Patient experience performance varied substantially by domain: discharge information had the highest statewide median at 86%, while medicine communication had the lowest at 61%. Patient recommendation showed a moderate negative relationship with readmission ($r = -0.301$, $n = 250$), and higher recommendation quartiles generally had lower average readmission rates. Ownership categories also showed descriptive differences in HCAHPS performance, although these comparisons do not establish an ownership effect.

● How will these insights help answer your business questions?

-In order to identify the hospitals and performance areas that appear to require the most attention, these outputs help the healthcare quality collaborative compare ownership categories, understand statewide performance distribution, identify hospitals with multiple screening concerns, separate low performance from incomplete reporting, identify weaker patient experience domains, and quantify unfavorable readmission performance.

Key tasks

- Aggregate your data so it's useful and accessible.
- Organize and format your data.
- Perform calculations.
- Document your calculations to keep track of your analysis steps.
- Identify trends and relationships.

Deliverable

● A summary of your analysis

-Population and data coverage: The final analytical population contained 285 Texas acute care hospitals. Overall CMS ratings were available for 200 hospitals (70.2%), selected HCAHPS measures were available for 256 hospitals (89.8%), and hospital wide readmission data was available for 254 hospitals (89.1%). A total of 199 hospitals (69.8%) had complete information for all three screening indicators.

-Overall hospital ratings: Among the 200 hospitals with a reportable overall rating, the mean rating was 3.36 stars, and the median was 3 stars. 42 hospitals (21.0%) received one or two stars, while 94 hospitals (47.0%) received four or five stars. The most common category was four stars, representing 69 hospitals (34.5%).

-Patient experience: The statewide median percentage of patients who would definitely recommend their hospital was 71%, while the median percentage rating their hospital nine or ten was 72%. Among the six selected HCAHPS measures, Discharge Information had the highest median at 86%, while Medicine Communication had the lowest at 61%. Definitely

Recommend showed the greatest hospital level variation, with an interquartile range of 13 percentage points and a population standard deviation of 9.32.

-Readmission performance: The median hospital wide readmission rate was 14.9%. Among the 254 hospitals with a CMS comparison result, 7 hospitals (2.8%) were classified as Better than national, 237 (93.3%) as No Different, and 10 (3.9%) as Worse.

-Ownership comparisons: Patient experience results differed descriptively across ownership categories. Among groups containing at least 10 hospitals, physician owned hospitals had the highest average results on five of the six selected HCAHPS measures, including a 77.8% Definitely Recommend rate, while proprietary hospitals had the lowest averages across all six measures, including 69.2% Definitely Recommend. These unadjusted differences should be interpreted cautiously because ownership groups may differ in hospital characteristics, service mix, location, and patient populations.

-Patient experience and readmission relationship: Patient recommendation and hospital wide readmission showed a moderate negative relationship ($r = -0.301$, $n = 250$). Hospitals in the lowest recommendation quartile had an average readmission rate of 15.15%, compared with 14.77% among hospitals in the highest quartile. The percentage classified Worse than national was 7.14% in the lowest quartile and 1.82% in the highest quartile. The quartile pattern and correlation indicate a modest association between stronger patient recommendation and lower readmission, but they do not establish causation.

-Screening results: Of the 199 hospitals with complete screening data, 3 hospitals (1.5%) met all three screening conditions and were classified as Priority 1, while 34 hospitals (17.1%) met two conditions and were classified as Priority 2. In total, 37 hospitals (18.6%) met at least two conditions. The screening conditions were a one- or two-star overall CMS rating, a Definitely Recommend percentage below the Texas median of 71%, and a CMS readmission comparison classified as Worse. The screening framework is intended as a transparent prioritization aid and is not a validated composite quality score.

-Key analytical limitation: Publicly reported CMS measures differ in eligibility and data availability, and 30.2% of the hospital population lacked at least one of the three screening inputs. In addition, the CMS overall hospital rating already incorporates patient experience and readmission domains, so relationships involving the overall rating are not statistically independent. The analysis is observational and should be used to identify hospitals and performance areas for further investigation rather than to infer causation or create a definitive ranking of hospital quality.

Share

Case Study Roadmap – Share

Guiding questions

- Were you able to answer the business question?

-Yes, with a qualification. The original question was: Which Texas acute care hospitals show the greatest opportunities for improvement based on overall hospital ratings, patient experience, and hospital wide readmission performance? The business question was answered well enough to identify a focused group for further quality review. The qualification is that the screening framework is not a validated composite quality score or hospital ranking. It identifies multiple transparent signals that warrant investigation.

- What story does your data tell?

-Texas acute care hospitals show mixed performance across quality domains. Nearly half of rated hospitals received four or five stars, but medicine communication was comparatively weak statewide, readmission results were heavily concentrated around national performance, and a smaller group of hospitals exhibited multiple overlapping concerns. Incomplete reporting also limits full comparison for roughly 30% of hospitals.

- How do your findings relate to your original question?

-The project moves from statewide benchmarking to relationships to prioritization. The analysis identifies: where statewide performance is comparatively strong or weak, which hospitals show several concerns simultaneously, which hospital groups contain a greater concentration of screening concerns, which measures may deserve further investigation, and where incomplete CMS reporting limits decision confidence.

- Who is your audience? What is the best way to communicate with them?

-The primary audience is the Texas Hospital Quality Collaborative's executive and quality improvement leadership. The best way to communicate with them is using a workbook for detailed analytical evidence and presentation for the small subset of visuals that tells the decision-making story.

- Can data visualization help you share your findings?

-Yes.

- Is your presentation accessible to your audience?

-Yes.

Key tasks

- Determine the best way to share your findings.

-See cms-tx-hospital-quality-presentation.pdf.

- Create effective data visualizations.

- Present your findings.

- Ensure your work is accessible.

Deliverable

- Supporting visualizations and key findings

-See cms-tx-hospital-quality-excel.xlsx.

Act

Case Study Roadmap – Act

Guiding questions

- What is your final conclusion based on your analysis?

-Texas acute care hospitals show generally mixed to positive performance across the CMS measures analyzed, but the results also reveal specific opportunities for targeted quality improvement review. Nearly half of hospitals with an overall CMS rating received four or five stars, while medicine communication was the weakest statewide HCAHPS measure and a small group of hospitals showed overlapping concerns across overall quality, patient recommendation, and readmission performance. The screening framework identified 37 hospitals with at least two improvement signals, including three hospitals meeting all three conditions. However, 30.2% of hospitals lacked at least one screening input, and the results should be used as a prioritization tool for further investigation rather than as a definitive ranking of hospital quality.

- How could your team and business apply your insights?

-The Texas Hospital Quality Collaborative could use the screening framework as an initial prioritization tool for allocating quality review resources. Hospitals meeting multiple screening conditions could receive additional review to determine which quality domains warrant intervention, while statewide HCAHPS findings could help identify common improvement topics such as medicine communication. The organization could also use the analysis to distinguish performance concerns from reporting limitations and to determine where additional data is required before making decisions.

- What next steps would you or your stakeholders take based on your findings?

-In this order, I would validate the screening list, review component measures, investigate priority 1 hospitals, explore medicine communication, investigate proprietary hospital differences, improve data completeness monitoring, and refresh the workbook periodically.

- Is there additional data you could use to expand on your findings?

-Yes. I could use many such as a historical CMS to determine whether flagged hospitals have had recent declines/improvements, hospital size and utilization to determine whether performance patterns vary between large/small hospitals and high/low volume facilities, urban/rural classification to determine whether missing ratings or performance concerns cluster geographically, and socioeconomic context to explore whether community

conditions relate to hospital outcomes or patient experience.

Key tasks

- Create your portfolio.
- Add your case study.
- Practice presenting your case study to a friend or family member.

Deliverable

- Your top high-level insights based on your analysis

-Nearly one third of hospitals lacked at least one screening input, primarily because overall CMS rating coverage was substantially lower than HCAHPS and readmission coverage.

Hospitals with incomplete data should therefore remain separate from hospitals with complete data and no flagged signals.

-Texas hospital ratings were concentrated primarily in the three- and four-star categories. Nearly half of rated hospitals received four or five stars, while approximately one in five received one or two stars.

-Medicine Communication had the lowest statewide HCAHPS median at 61%, compared with 86% for Discharge Information. This makes communication about medicines an appropriate statewide area for further quality-improvement review.

-More than 93% of hospitals with a reportable readmission comparison were classified as No Different from national performance. Only 10 hospitals were classified as Worse and seven as Better, meaning this measure is most useful as a targeted screening signal rather than a broad hospital ranking tool.

-Patient recommendation showed a moderate negative relationship with hospital wide readmission ($r = -0.301$, $n = 250$). Hospitals in higher recommendation quartiles generally had somewhat lower readmission rates, but the observational relationship does not establish causation.

-37 hospitals met at least two screening conditions, including 3 hospitals that met all three. This provides the collaborative with a focused group for additional review while retaining the transparency of the individual measures used to identify them.

-Proprietary hospitals had the greatest concentration of multiple screening signals: 27 of 113 (23.9%) were Priority 1 or Priority 2, including all three Priority 1 hospitals. This descriptive concentration warrants further investigation, but the analysis cannot establish that ownership itself explains the observed differences.

- Based on what you discover, a list of additional deliverables you think would be helpful to include for further exploration

-It would be helpful to include many things like a multi-year hospital trend dashboard, hospital peer comparison report, data completeness monitoring report, and geographic performance map.

